

Dirichlet Characters

For each of the following descriptions of a Dirichlet character of modulus $72 = 2^3 \cdot 3^2$ determine the conductor.

In each case explain how you arrive at your answer.

Question 1

A Dirichlet character χ of modulus 72 such that its image is precisely $\{0, 1, \omega, \omega^2\}$ where ω is a cube root of unity.

Solution. We write the character as a product $\chi_a \cdot \chi_b$ where χ_a has modulus $8 = 2^3$ and χ_b has modulus $9 = 3^2$.

A character of conductor 4 or 8 has to take the values ± 1 since

$$\left(\frac{\mathbb{Z}}{\langle 8 \rangle}\right)^* \cong \left(\frac{\mathbb{Z}}{\langle 2 \rangle}\right)^2$$

Thus, χ_a is $\chi_{0,2}$.

We note that

$$\left(\frac{\mathbb{Z}}{\langle 3 \rangle}\right)^* \cong \frac{\mathbb{Z}}{\langle 2 \rangle}$$

Thus, if ω is in the image of χ_b , then χ_b *cannot* be of modulus 3. In other words, χ_b is of modulus 9.

It follows that χ has modulus 9.

Question 2

A Dirichlet character χ of modulus 72 which takes values in $\{0, \pm 1\}$ where $\chi(5) = 1$ and $\chi(17) = -1$.

Solution. As above we write $\chi = \chi_a \cdot \chi_b$ where χ_a has modulus $8 = 2^3$ and χ_b has modulus $9 = 3^2$.

A character with conductor 9 must take 4 to an element of $\{\pm\omega, \pm\omega^2\}$. Since the image of χ is contained in $\{\pm 1\}$, we see that χ_b has conductor at most 3.

Now $\chi_a(17) = \chi_a(1) = 1$. Thus

$$\chi_b(2) = \chi_b(17) = \chi_a(17)\chi_b(17) = \chi(17) = -1$$

Thus, χ_b is *not* the principal character $\chi_{0,3}$. Hence χ_b has conductor 3.

Using $\chi_b(5) = \chi_b(2) = -1$, we calculate

$$\chi_a(5) = -\chi_a(5)\chi_b(5) = -\chi(5) = -1 \neq \chi_a(1)$$

so that χ_a does not have modulus 4. Hence, χ_a has conductor 8.

Thus, we see that χ has conductor 24.

Question 3

A non-principal Dirichlet character χ of modulus 72 which takes values in $\{0, \pm 1\}$ where $\chi(5) = \chi(17) = 1$.

Solution. As above we write $\chi = \chi_a \cdot \chi_b$ where χ_a has modulus $8 = 2^3$ and χ_b has modulus $9 = 3^2$.

A character with conductor 9 must take 4 to an element of $\{\pm\omega, \pm\omega^2\}$. Since the image of χ is contained in $\{\pm 1\}$, we see that χ_b has conductor at most 3.

Now $\chi_a(17) = \chi_a(1) = 1$. Thus

$$\chi_b(2) = \chi_b(17) = \chi_a(17)\chi_b(17) = \chi(17) = 1$$

Thus, χ_b is the principal character $\chi_{0,3}$.

Now, we calculate

$$\chi_a(5) = \chi_a(5)\chi_b(5) = \chi(5) = 1 = \chi_a(1)$$

so that χ_a has modulus 4. Since χ is non-principal, χ_a must be the *unique* non-principal character of modulus 4 such that $\chi_a(3) = -1$.

Thus, we see that χ has conductor 4.